

Requirements Questionnaire

Project Title: AI-Powered Summary Generation for Doctor

Group Members:

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1. General Project Questions

- Who will be using the system (doctors, assistants, others)?
Answer: Doctors are the primary users; assistants may use it for testing.
- What will the system be primarily used for?
Answer: Generate AI-assisted summaries from clinical notes (typed and possibly handwritten).
- What platform(s) should the system run on (desktop, mobile, Windows, Mac, web)?
Answer: Web application, accessible via desktop browser (Windows/Mac).
- Should the system support multiple users (e.g., different doctor logins)?
Answer: Yes, multiple doctor logins with simple role handling.
- How many users (doctors, assistants, clinics) do you expect to use the system initially and at full scale?
Answer: Prototype will support 1–2 doctors; full scale (future) could support multiple clinics.
- Should different roles (doctor, assistant, admin) have different levels of access or permissions?
Answer: Doctor (full access) and Admin (system setup); assistants not required for first version.
- At what point in the workflow (during visit, immediately after, end of day) do you expect the system to be used?
Answer: Mainly after the visit (batch or end-of-day summarization).

2. Functional Requirements

- Should the system support both handwritten and typed notes?
Answer: Start with typed notes; stretch goal is basic OCR for handwritten notes.

- Should it be able to process multiple files/entries at once?
Answer: Support multiple files uploaded at once.
- Should the system store full patient files, just the redacted summaries, or neither (due to HIPAA)?
Answer: Store only redacted summaries (no raw PHI).
- Should summaries be downloadable/printable for records?
Answer: Summaries should be exportable as PDF.
- Should summaries be editable after AI generation?
Answer: Yes, doctor can edit after generation.
- Would you want this tool to export summaries for use in other systems in the future?
Answer: Prototype will not integrate directly with EHRs; may allow JSON export as future step.
- Should the summaries include references or citations back to the original text?
Answer: Yes, show highlighted source text to improve trust.
- Do you need integration with existing systems such as EHRs (e.g., Epic, Cerner) or coding standards (ICD-10, SNOMED, CPT)?
Answer: Out of scope for the time; placeholders only.
- Should user corrections to summaries feed back into the system to improve future outputs?
Answer: Edits won't retrain the model; captured feedback may be logged for demo purposes.
- In what formats should summaries be exportable (e.g., PDF, Word, JSON, FHIR resources)?
Answer: Export as PDF and Word in v1; JSON optional stretch goal.

3. Non-Functional Requirements

- How simplistic should the UI be?
Answer: Minimal, clean interface; single-page upload + output view.
- Is there a specific scheme we should achieve for the UI?
Answer: Basic Bootstrap/Material design, no branding.
- How advanced should the security be (encryption standards, MFA)?
Answer: Basic (password auth, HTTPS); MFA not required in prototype.
- If login fails, should password reset be supported?
Answer: Yes, via email link or mock workflow.
- What is the expected performance (time to generate a summary)?
Answer: Target ≤ 10 seconds for a one-page typed note.

- In the event the system isn't working properly, what backup or workaround would you expect?
Answer: If AI fails, display original note with placeholder summary.
- Should the system maintain audit logs (who accessed, viewed, or edited summaries)?
Answer: Log who logged in and generated summaries (basic).
- What uptime or availability targets do you expect (e.g., 99.9%)?
Answer: Not relevant for prototype; local hosting acceptable.
- Should the system be accessible to users with disabilities (WCAG compliance, screen readers, font scaling)?
Answer: Follow basic accessibility (clear fonts, alt text); full WCAG out of scope.
- Do you need support for multiple languages (e.g., Spanish as well as English)?
Answer: English only; Spanish may be a future extension.

4. Constraints and Assumptions

- Should patient data be stored locally, in the cloud, or not at all (privacy-first)?
Answer: Local server (Marist machine or cloud dev instance); no long-term storage.
- If storage is required, how long should information be kept (temporary vs. long-term)?
Answer: Temporary (data removed after session or logout).
- Who should be allowed to access stored data (only the doctor, assistants, multiple clinics)?
Answer: Only authenticated doctor users.
- Are there budget or resource limits that constrain technology choices?
Answer: Free/open-source tools; free cloud tiers; no paid APIs unless pre-approved.
- Are there institutional or legal compliance requirements we must follow (e.g., HIPAA)?
Answer: Prototype will simulate HIPAA awareness but not achieve full compliance.
- Do you expect this system to function as a prototype for demonstration purposes, or as a real deployable tool?
Answer: Strictly prototype/demonstration.
- Are there institutional IT constraints (e.g., must run on-premises, inside a secure VPC, or with a specific vendor)?
Answer: No need for on-prem VPC; simple web app is fine.
- Are there constraints on hardware resources (e.g., must avoid GPUs, or can use cloud GPUs)?
Answer: Use local dev laptops; free cloud compute if available.
- Who will own the data and models generated by the system (institution, vendor, or shared)?
Answer: Data and code belong to course/Marist.

- Are there licensing or intellectual property concerns we need to plan for?
Answer: No proprietary libraries beyond what's allowed in class.

5. Risks and Dependencies

- Should the system alert users if HIPAA compliance is compromised?
Answer: Not implemented; only warning text in UI.
- How should the system theoretically handle data breaches?
Answer: Low risk; prototype will avoid storing sensitive PHI.
- What concerns do you have about using AI for summarization? (e.g., leaving out important details, making mistakes, privacy issues)
Answer: Possible omission of details, inaccuracies, privacy.
- Should the system be able to work without internet access, or is it acceptable if it relies on cloud services?
Answer: Internet required (cloud model API).
- Are there restrictions on sending sensitive data to third-party services for processing?
Answer: Use only de-identified or synthetic data; avoid real PHI.
- Should the system check for potential bias (e.g., performance differences across specialties, accents, or note types)?
Answer: Not feasible in timeframe; acknowledged risk.
- How should liability be handled if the AI omits or misrepresents critical information?
Answer: Prototype only; outputs not for clinical use.
- Are there dependencies on external vendors (OCR services, APIs, EHR integrations) that could introduce risk?
Answer: Cloud AI API (e.g., OpenAI), OCR library (Tesseract), web framework.
- Should the system flag uncertainty in its outputs (e.g., highlight sections it is less confident about)?
Answer: Highlight sentences where confidence is low if API supports it; otherwise, note as limitation.

6. Validation and Prioritization

- How will we know the summaries are accurate and useful enough for real-world use?
Answer: Compare AI summaries with 'gold' provided notes (see attachment); evaluate coverage of diagnoses/meds.

- Which features are essential for the first version, and which are nice-to-have for later?
Answer: Essential: Typed note summarization, editable output, PDF export. Nice-to-have: OCR, JSON export, source text highlights.
- What level of accuracy is required for clinical adoption (e.g., percentage recall/precision for diagnoses, medications, allergies)?
Answer: For prototype, aim for $\geq 80\%$ recall of key terms (diagnoses, meds, allergies).
- Should the system be piloted with a limited number of users or notes before full deployment?
Answer: Internal testing with sample notes; no real clinical pilot.
- How will user satisfaction be measured (e.g., surveys, SUS/NPS scores)?
Answer: Informal survey from client/peers.
- Should certain clinical elements (e.g., allergies, medications) be prioritized for higher accuracy than others?
Answer: Allergies and medications most critical; others secondary.